ActInf Livestream #028.2 ~ "Towards a computational phenomenology of r

Livestream | 1:54:12

hello and welcome everyone to the active
inference lab this is actin flab
live stream number 28.2
on september 14 2021
and we're going to be having our dot 2
follow up discussion on the paper
towards the computational phenomenology
of mental action
modeling meta awareness and attentional
control with deep parametric active
inference
welcome to the active inference lab
we are a participatory online lab that
means you can participate that is
communicating learning and practicing
applied active inference you can find us
at the links here on this slide
this is recorded in an archived live
stream so please provide us with
feedback so that we can be improving our
work
all backgrounds and perspectives are
welcome here
we'll be following good etiquette for
live streams and see who joins in
at this short link you can find the
calendar of events for 2021 as well as
all the other kinds of streams that we
do
related to kind of wild card guest
streams
and the more technically oriented model
streams and math streams
and we have some preliminary decisions
for what we'll read in 29

today though we're here to continue the
discussion on 28
paper which is the towards a
computational phenomenology of mental
action paper
and again we'll see who joins and see
where we go i wrote down a few comments
and questions but i know that there's
many ways it can go and of course
everybody who's watching live definitely
um
leave us questions and comments and we
can address them on the stream
uh
everyone's participating who's here at
this time
we can start with just some
introductions and warm-ups since there's
only a few of us we can
say hi and um
reminisce on what brought us to this
paper or even where we'd like to go
for the dot too where are we
jumping off from or two as we're in that
kind of the middle of the gangplank the
middle of the diving board so
i'm daniel and i'm a researcher in
california
and i think today i wrote down the
pieces that i wanted to
if not resolve just at least explore a
little bit and it had to do with
annotating that generalized model of
mental action and seeing if we could
find words in a language or scenarios or
exercises thought experiments
find some tangible ways to interact with
these different aspects of uncertainty

and then
on a more theoretical note
ask how are those maybe related to
neurological function or certain kinds
of measurements that are made in the
laboratory but just at the experiential
level
are there cases where we actually can
get in touch with some of these
different
qualitatively different kinds of
uncertainty
so
either or stephen just go ahead stephen
avendy
yeah good morning
from toronto um
yeah i'm interested
to sort of pursue what we're talking
about in the last session and
i'm very interested in the fact that
a bit like um to some extent
process-orientated
psychology
by um the min dales which comes out of
some work around information theory in
the 70s this idea of really getting into
the process from within
and the phenomenology so this this this
paper starts to look at
giving
actual models or at least a heuristic as
well for thinking about how we
phenomenologically can be doing stuff
from the inside and for me that's
exciting because
psychology can be kind of limiting
because it's always this external

modeling
of people which then sort of but how can
we get and look at some of the
structures of how we
operate at some level so this this is
cool and i think this is the first
or one of the first papers that really
gets into that phenomenology beyond a
philosophical point of view and starts
to sort of
give a few handles on it and
that's where i'm interested so i'm going
to pass this over if it's okay dean
hey good morning my name is dean i'm in
calgary
and a few years back i read a book by
a philosopher by the name of douglas
hofstetter it's called i'm a strange
loop
and
yeah sort of trying to put some sort of
a
context on when i read that book and
then this paper
um
hofstetter doesn't really get into the
um
the perception of
i'm aware of planning to make a plan
because i can think about my thinking so
what have this planned policy of
attention and then all of the math that
supports that
um
and then he sort of doesn't really get
into
and then what i attend to tells me what
i'm losing focus on which is what this

paper also talks to so again i'm just
trying to i'm trying to put these two
ideas side by each
and without saying
either one of them is more correct i
think they're very complimentary so
yeah that's where i kind of was
looking at this and saying hmm
this is really interesting so yeah
that's where i i'm kind of coming out
this room so
on the hofstetter the strange loop and
the geb some works that i really love as
well as mind's eye with
dennit
yeah it does seem like his focus is on
how symbols
are computed upon which makes sense
given his kind of computational focus
and so we've talked a lot about how
syntax and semantic layers can be
seen within active inference and
that doesn't have phenomenology
in the picture at all
and that's why the sort of question
about how computers interpret syntax and
how that becomes semantics
is orthogonal it's kind of you could
take one take on another
whether computers have experience or not
like you don't need to think that
computers are having an experience or
not to ask how a programming language is
compiled or interpreted so there's sort
of that two by two with how meaning is
made in a computational sense
and then
how our meaning is made and experienced

uh so that's definitely an interesting
piece
um and steven you wrote like from the
inside what do you think differentiated
this
approach
rather than the theoretical or the
philosophical and what does it look like
for an academic
work or for a work of research to
not just be theoretical or philosophical
yeah this is a good question um
well the i suppose
having sort of um
mechanisms and
identifiable relationships between
aspects of experience
i.e
thinking about thinking and
our
sense of
accuracy in our predictions and things
like that gives a way for stuff to
relate to each other otherwise it tends
to be
one
thread of that so be it consciousness be
it um
awareness
be it that one thread then can be mused
about and we could get quite an
interesting philosophical explanation of
what it is
the thing is though that
you can't really process that or try to
emulate the process of that on its own
there needs to be something else that is
somehow

in like you say the body
what's the body you know
some sort of sensory feelings
um some sort of sense of predictions
about something else and
normally that seems to be where things
fall down a bit ends up going off to the
neuroscience world and
again which is great but still ends up
being a kind of a model of the brain
right so
you're in this neurophenomenology world
where it's kind of explaining
phenomenology but through the brain but
again i can't the brain has got no
feelings inside it you know
so you can't be your brain i think we
think we can because we have all these
pictures of brains as if they are people
but
um so i think
it's interesting questions i don't
really quite thought about it this way
but i think there's something about
having different aspects of our
phenomenological
um being able to
meaningfully relate
um
and not end up being kind of a deep
explanation of one strand i don't know
if that makes sense
it also makes me think about in
hofstetter's girdle usher block geb
one of the main strange attractors
is the turing test so that's the
question about how you can identify on
the other side of the blanket some other

system in another room
how do you know whether it's
a human or whether it's a computer or
whether it's aware or not so
there's been all kinds of thought on
that
and
it's almost like
what happens when you take that
as a fundamental unknowable
how do we respect what's on the other
side of the wall
and maybe even give it an affective
parameter
while recognizing that we have the same
and we're experiencing it but we're
never going to be able to experience
what's on the other side of the wall
we're going to be only able to
experience our interactions with what's
coming across
and i think that's why
hofstetter and dennett worked well
together
because then it with the intentional
stance is kind of like saying
it's simple it doesn't really matter if
you're playing chess it doesn't matter
if you're playing somebody who's human
or a super computer or something it's
about what's on the board and in a way
that's kind of like the way that we're
addressing it's about the sensory input
coming in and how we're modeling the
external states so if it's in the other
room the turing test is
on the outside of our blanket we're
experiencing what's inside and that

could be an illusion or it could be
super real or all that's real depending
on again which philosophy you decide to
lean towards but
the question about how we interpret and
experience our own
parameterization of these models is
really secondary from the implications
of what's on the other side of the veil
steven
yeah that's a really useful um
lens to bring to play actually and
actually the game brings this back out
maybe so that we're not suddenly focused
only on the individual
isolated from the world
but one thing i think is really
interesting with that is
with active inference there's this
well n active inference we're going to
go down that route is that you know you
have the
you know with your body right so
ultimately
the active agent can check in
and i think what you're just saying is
one thing that i would know over a
period of time if something was alive is
if they
had to reflect on
some aspect of our discourse okay
and it was something which neither i
could know or they could know
i
but when they checked in with themselves
they came up with an answer which i
could find plausible that makes sense so
there's no way that you you know you

know there's those things that what do
you feel about this well i feel and then
you come up with something he's like we
there's a knowing that you come up with
right but that wasn't in your semantic
lexicon of pre-programmed stuff you were
able to check in somehow
and that's the sort of thing an
artificial machine
can't do unless it can check into some
some way of knowing because
if it wasn't known before how could it
have ever been programmed into the
supercomputer
so that's kind of interesting
it comes up in this paper with a
question about transparency and opacity
so again the kind of what is water
transparent states are the ones that you
see right through so you actually don't
perceive them directly like air and then
the opaque states are the ones kind of
like you know the matte black wall you
can see it really well and so this
question about how a prompt
can almost like a phase transition
move something from being transparent to
opaque like how does your leg feel right
now
and it moves it from being a transparent
state potentially one that attention is
not being paid to
to an opaque state which is something
that attention is being paid to that's
one of the major contributions of this
paper which is to take metzinger's
framework of transparent and opaque
states

as applied to phenomenology
and bring that into the
affective framework
developed by hesp at all for the deep
parametric active inference so it is
like these multiple threads coming
together and then
it's just so appropriate that attention
is kind of the thread that links it and
precision and awareness are kind of
coming together in new ways
[Music]
um the two things i wrote down and of
course anyone watching live definitely
ask random questions no off topic
questions
no such thing exist not not forbidding
such questions
um so one thing as kind of addressed
earlier was i thought it'd be fun to
annotate figure 12 a bit and explore
some of these qualitatively different
forms of uncertainty
and
but maybe a lead-in topic and again
anyone can just
feel differently or ask a question a
leading topic was like
how do we think about uncertainty
greetings fellow jitsu blue jitzer
uncertainty in remote work just
how often when you send a message
you don't get that instantaneous
feedback that you would
in
a video conversation or in an in-person
conversation because the time lag is
very different so it's almost like

you're left hanging
with the uncertainty or you have to come
to a new way of understanding
your uncertainty because you're not
or you're expecting not to hear from
them oh i know that they're on the other
side of the world so i'm expecting not
to hear from them for the next hours
just
i wondered what sorts of these um
personal level
uncertainties applied to the remote case
and then thinking about the remote team
as its own active inference entity
are there uncertainties that exist
for located or remote teams
that
don't necessarily
find themselves simply explained by
personal level affect
are the things that teams are uncertain
about that people are certain about like
a hung jury
or a split election or decision
are there things that people are
uncertain about but that groups can be
quite certain
[Music]
about you
but that's just things that we can
return to in any
time
just got my package through for my mum
for baby clothes from england so i'm
back nice
nice
um
[Music]

i think blue wants to say something yes
blue
uh yes so just to that point i mean i
think that like that's like the
the kind of like lifeline or ask the
audience like in the you know when you
have uncertainty about a question i mean
an individual can be uncertain but
the um crowd can have some certainty and
wasn't there something about guessing
the weight of a bison or
some story like that like where you know
the the i think it was the median value
was exactly the weight of the bison like
from this entire like divergent crowd um
i'm not sure the exact i don't remember
like the exact story it's just kind of
faint in my memory but i mean i think
that definitely the crowd knows what
the individual can't
maybe there's many oh yeah steven go for
it
yeah well that yeah that's an
interesting one that um
blue mentioned and and also i suppose
there's this question well let's say we
take uh you know bayesian statistics you
know if it's a distributed sense of
something
well so when you've got a very large
number there may be this distributed
sense which as if people have got a feel
for it
i mean i suppose it assumes that there's
not complete noise there's the people
have got some kind of idea out of that
group but there's
some knowing but in a team and stuff

there is this question and this may be
really what types of things we're
evaluating is with a team
or a group
because teams tend to have a particular
teleology if we're trying to get a
handle on something intersubjective
i mean a classic case with this would be
saying in a marketing agency or
something like that or with a film
you know you might want to
get a sense of
you might get market research but you
also want the creative team might get a
sense of
what is um you know what's what how's
this going to land you know what's and
that that might be an integrated thing
um and it might be the other creative
director that
is pushing out their one vision because
they're trying to be novel they're
trying to give something that breaks
the group
um expectations
however you also then want to
stay within i mean a classic one would
be like ricky gervais before the um when
he did his uh what was it the it's not
the oscars but he did the other one the
golden golden globes
at some point he had to kind of wait
what he could get away with right
so there would have been a whole load of
people sort of
seeing how that could get tweaked and
nudged so that
it basically fits on that

very near edge and that's uh
you know that that would tie quite well
with the fact that you need to have more
you can't it's not just multi-sensory
integration on one person that's enough
you might need
multi-inter-subjective
awareness integration to get a better
sort of integration
right you need to know how others
thinking through other minds or theory
of mind you need to
kind of have an estimate of how others
will
feel about your stimuli from an
affective dimension
and
blue i think
the example with the ox and guessing the
weight it's uh francis skalton sir
francis galton perhaps
and i just looked it up it was a 787
fair goers
who predicted the weight of this animal
and that the average was extremely close
even though of course many individuals
probably
were over or under by vast margins and
it ties us back to the strange loop and
aunt hillary the ant colony in geb
and when does collective decision making
work collective decision making amongst
cortical neurons it seems to work pretty
well that's kind of what evolution has
shaped
why did it work for predicting the
weight of that ox well potentially many
people were familiar with how much

animals weighed
another answer is that it was within an
order of magnitude like you could
probably say well it's probably not 100
pounds and it's probably not 10 000
pounds
so when there's bounds to the estimate
rather than it being distributed over
like many orders of magnitude there's
a case for the average being something
that can be taken and that actually has
analogy to a lot of these mean-seeking
distributional techniques that we
utilize like the case where the actual
distribution might be very rough
but we use variational techniques
that can pursue for example the mean
of the distribution and just sit the
weight of our distribution that we
control q
and sit
with only one or a few parameters
right over the bulk of the distribution
rather than getting into the hundred
parameters that might describe the
actual underlying distribution
and so it's like
under what cases
can distributions be well approximated
by their mean and by many independent
estimates
and under what cases might you be
looking just like steven said with a
marketing agent
trying to be creative or novel
under what cases are you actually
looking not to estimate the mean
but rather to propose a counter factual

or to make something that is going to
succeed by virtue of its novelty
stephen
yeah and also it takes you into this
world of the of the rural community i
don't know if it's rural america you
know
they will have their wisdom about those
types of things their fair go as they
they probably got quite an insight into
animals you know they
i bet you could get a fairly good idea
of how slippery it would be to try and
grab hold of that baby hog you know in a
way that the city folks wouldn't have a
clue so it's a
this kind of contextual
social like
knowing um it's quite you know that that
that i think that's a good thing to
to bring in as well when we think about
participation you know then this knowing
that can be in different communities
outside of the
the cities which you know and the
advertising agencies and these
dominating kind of
media
sort of
leaders of thought
i think it shows how we need to be able
to bring in
these more disparate ways of
experiencing the world
well when it seemed that there's a mean
or an average like there was the famous
like um like average american survey or
something like the dimensions of the

average person but then like no single
person
was within a tolerable limit of that
average yet there was an approximation
um versus
casting it into the experience like how
will this be received by different
people it includes in the question
the need to think through other minds
and um
that no one was thinking through the
animal when they had to predict the
weight that's that external the third
person psychology and it was estimating
a physical attribute so it's not like
they were trying to do anything
different but now if you asked on a
scale of one to ten
how do we estimate how happy this animal
is
the variance around that estimate is
going to be of a very different kind
and potentially an irreducible kind
we're not just looking for a point
estimate to converge on
maybe there's differences in opinion and
that is something that would be revealed
through that question
whereas
if you're wrong about the weight it's
just because maybe you're
not used to it or something like that
steven yeah
yeah that's a good that's a good point
to bring in as well is when looking and
say also we maybe want to bring in this
bayesian inference as well like if you
could have multiple rounds you know like

when carl fristan does some of the
demonstrations
it would be interesting
a to be able to have some way of looking
at something and using multiple
iterative rounds which they do in
a lot of processes like future search
and um
other kind of
group participatory processes and also
though
to break apart the generative models at
play so you and this is this is the
challenge is with a lot of things is
with normal statistics is you'll get an
answer you'll get a graph if you just
take all the numbers and divide by x
right
and then so someone's now got research
results but the thing is like you say it
makes a difference if you've got three
models two models five models one model
going on of the reality that they're
trying to in theory that that the weight
of the cow there's
probably one model there might be a
couple of different ways of trying to
think about how to go to that model but
you know this is sort of a fairly clear
proposition but
if you do have more than one generative
model or one contextual
engagement that's being used teasing
that out and then having a way to
iteratively
do beijing inference and active
inference
on each of those models so they can be

teased apart would look like it'd be a
useful thing
cool dean
yeah i'm just going to say to steven's
point there's a
book called the end of average by todd
rose
that speaks to those times when
finding the mean doesn't really work and
you do have to kind of pull that
generative model apart
so if you're if you're interested in
that there's there's a great reference
thanks dean
and then um this idea of updating
it's almost like we have a vertical
updating which we're gonna look at when
we look at the deep parametric model
it's an updating within
the agent within the system passing up
and down top top down bottom up but it's
an updating inside of the agent versus
the updating laterally or across the
ensemble of similar agents like imagine
if the people instead of putting their
weight estimate for the for the ox into
a box
they actually talked about it
there's cases where you could imagine
that would actually
make a worse estimate like a really
persuasive person is convinced they have
high precision about their estimate
they're wrong but they have high
precision and then in that case it's
possible for them to shape group
outcomes
like a more stubborn peg

whereas there's other cases where when
maybe um it's less important to be
precise on a point estimate but like to
brainstorm maybe sharing ideas early on
helps increase that temperature and
updates people's models in a way where
just having them in separate lanes and
then coming together to converge at the
end it's it's too late for those ideas
to be meaningfully compared because
they're of such different type whereas
again if everybody is constrained to
we're fitting something about weight
only then maybe it's okay to combine
them later on in the process blue
so i was trying to flip the paper just
now but i think that they've looked at
that actually like when you have really
strong
conviction about your
beliefs or ideas they look at how
information is spread in networks and
really like those people that are
you know very strong in conviction they
like contaminate
their like neighbors in the graph
essentially so it's really neat to kind
of see this like information propagation
and how it really does start with the um
the people with the strongest
convictions
it's almost like uh yeah yes people are
spreading information based upon a few
things but up there on the list would be
how precise they are about it
and also how they want to be seen
so both of those are like first and
second level estimates of precision

about oneself metacognition i won't
raise my hand if i'm not totally sure or
you know i'm watching a live stream but
i'm not sure if this is a good question
to ask hint it is but then you don't
know so you don't make the action
versus how will even other people
perceive it
that was probably less important in an
anonymous
guess about weights but as we talk about
issues that are increasingly social
those
become the drivers maybe 90 of people in
a room or in a country feel a certain
way but it still isn't perceived as
something that's okay to share
stephen
yeah this also ties into when
a lot of context that's actually a
mediator chapter was a mediator he had a
really good quote that he actually came
up with it was saying how
um
process both conscious and unconscious
implicit and explicit has a greater
impact on outcomes
than substance
i.e
so if you're going to bring people
together like we're saying and they're
going to try and work together being
able to
the process through which people
interact
can have a bigger impact on the outcome
than the substance partly because of
exactly what blue was saying

and you were saying you know around
whether someone becomes a dominant voice
whether certain information if it all
becomes about the substance it's very
easy to get trapped in this
who's right who's wrong well if in this
instance we're trying to get
a different handle on things
or to try and get a an active influence
we need that process to be given a
chance um
so i think that's quite and that's
actually also had a good conversation
yesterday with uh
um gary kirkham actually does a lot of
device theater work through so
interesting processes where they mix
different artists and one thing there is
when they go into the action space of
exploring
they would have the director watching or
they have like a device there to play
right watching
and in some ways the people who are in
it they just are totally immersed in it
they can't really reflect on where they
are at that moment it's just in there
and there's all this information and the
people on the outside are watching and
you could say i suppose they're actively
inferring on what might be easier to say
this bayesian
integration and waiting and then
afterwards they will be told what worked
what didn't work and then they could
bring that to their conscious awareness
and maybe
bring that in again with some but that

process
in a way needed to be just a living
process and not
too cognitive for the person to give
good quality
fuzzy information
so
uh two things there
the first is what do we say about active
inference it's a process theory
it's a corollary of free energy
principle and it probably draws on many
other domains as well but it's a process
theory it's not um a state theory and in
the alias 2018 interview fristen talks
about process versus state so that's a
lot like this process versus substance
you derived these are
a set of patterns
for how systems update
and so it is about the process of active
inferencing which i hear people using
sometimes which even differentiates it
even more from being just a static
substance theory like it's a static
theory about active entities it's like
kind of but it's actually a dynamic
theory
about static and dynamic entities
because you could have an active
inference agent
that is just a thermometer so its
affordances are nil so can't act but we
can still put it within this bayesian
graph framework it doesn't need to be
far from equilibrium it doesn't need to
be intelligent it doesn't need to be
actively foraging for information we

know the model can go all those places
but it's almost like with the process
theory you can explain active and
passive processes whereas with a
substance theory you're able to explain
static things pretty well but then
there's always this question about using
the static theories to apply to dynamic
systems
and then
also to
connect to the first piece of what you
said and to return to maybe some of the
threads in this paper you talked about
how that was a mediator
and so
mediation can look like so many things
from diplomacy
negotiation
maybe even marketing like you raised
earlier but mediator sounds a lot like
meditation
like the mediation is like we're we're
not just going to make a call and then
you're going to live with it it's like
we're going to mediate
now what are we mediating
when we meditate
are we mediating our thoughts and these
kind of mindfulness cues that lars was
providing last week
like putting the distance between
oneself and their thoughts in a way it's
like being a meta mediator or somehow
mediating
lower level
cognitive processes
through the process

not the substance that we call
meditation blue
so maybe like what are we mediating in
meditation that's like a super
interesting question and i wonder if
it's um
you know like the information
transmission in the brain or maybe
uh propagation or like the ongoing like
wandering mental wandering maybe so
there's
you know if we're if essentially
meditation
cuts that information transmission right
so so when you just are like letting go
let go of the thought right like the
thought arises just let go so the
thought doesn't keep wandering around
and i wonder if a thought spreads like i
don't know i mean i know my background's
neuroscience but i don't know enough
about
or maybe no one knows enough about the
brain and functionality to know like
does a thought propagate like from one
region of the brain to another region of
the brain like stimulate a new thought
right like and is that how like mental
wandering maybe works i mean i don't
know i mean it's just it's just
hypothetical but but maybe we're cutting
that information transmission process
off when we're mediating through
meditation
stephen
yeah it's interesting i thought blue
about tren it's actually funny because i
was just watching this video on process

orientated psychology and it was talking
about moving between modalities you know
like you know it could be you know i
have a thought
having seen something and then maybe it
processes into how i hear it
and how i see it and how i
color it you know like it could be you
know this it could be interesting how
the different sensory modalities could
um you know could hold a thought for a
period of time and then pass it back
again because synesthesia obviously is
an interesting one on that and i've just
seen an interesting paper actually that
um
talks about how
um
the vowels i-e-a-o-u
they mapped it out for colors and they
find that there's like a color map for
how people see it and that maps across
swedish and english i think it was
swedish in english
so
that's interesting as well that there
could be
you know these subconscious correlates
thanks steven blue
yeah so i mean i really don't know that
we know enough about a thought and like
this reminds me of um our guest stream
with shanna dobson like
what is the fundamental unit of a
thought right and and so i know in
a biological sense like the fundamental
unit of a thought really kind of has to
be the action potential but

like is it one action potential is it
like some critical number of action
potentials that instantiates the
thoughts so i don't know um and
definitely like there is that that
propagation or like a spreading some
contagion maybe to like other brain
regions in synesthesia i think that
that's been pretty well established like
that there's like some crossed wires and
so like you get auditory perception for
example in the visual cortex and so i
know that there's that relationship so
that kind of does like lend itself to
this idea of like the thought
propagating through different um areas
of the brain that but that's in in a
very incorrect sense i just wonder even
in normal people does a thought like
what is a thought how many action
potentials make up a thought and then
does that can that move around like
within a region or to different regions
is that actually what's happening i mean
i don't know that we can quantify that
at all but it's just it's interesting to
think about
not to always come back to aunt hillary
the ant colony in geb but that's the
analogy that hofsteder makes it says if
ensembles of neurons are transiently
activated and even for the same word
it's a different ensemble of neurons
which is increasingly what molecular
neuroscience is showing to be the case
and similarly with the ant colony it's
like the task group is assembling and
disassembling and reorganizing the idea

of a thought is still that noun centric
approach
which would lend us to want to find a
static representation versus like a
thinking unit or something that reminds
us that it's a process that thinking is
a process
um
so one comment and then a nice question
in the chat
about the mediator and the meditation so
let's go to figure five which is
again one of the
main contributions of the paper
so it's this three
level
nested structure
and
the outputs of one level
influence
a lower or an interior nested model so
on the bottom in blue
is the perceptive states like what am i
perceiving is it
light or dark
the second level is the attentional
layer so contrast this with for example
lstm
or other types of attention based neural
network
architectures which are not based upon
first principles
uh per se
and then what this paper adds in
is
not just the affective components
but actually this
precision of how aware am i of where my

attention is so then thinking about the
mediation and meditation i thought well
one mediation that might be happening in
active inference
is the mediation between epistemic and
pragmatic goals so that's kind of like a
seesaw and there's some sort of
negotiation happening
with free energy minimization that is
balancing it's mediating these two goals
when the two goals are perfectly aligned
great then it's like we agree
but when they don't perfectly align when
the pragmatical just wants to run up the
mountain and then the epistemic goal
wants to ask
should we look around and see if there's
another way that might be better in the
long run yeah can we go around exactly
dean solenoid yep the solenoid will flow
and so
there's sometimes a tension there and so
that's one type of mediation
but then
using the mediator more like the the
higher level diplomat here
it's almost like the green layer
is serving as a mediator of attention
by
calling awareness to it and this is
awareness of awareness that's why it's
metacognition
but it's mediating
the dynamics of the second layer which
end up cascading to influence the
perceptive layer as well
but
in a sense it is mediating

the activity of the second layer and
certainly this is the level that in the
paper they're saying is where meditation
rests that's why when we just had active
inference plus attention we didn't have
papers on meditation
but now that we have active inference
plus metacognition awareness of
attention
we bring in this whole experiential
dimension
with meditation stephen
one question i might have here
um
at that bottom and say what am i
perceiving so that's kind of fair enough
there's that kind of
what's
you know i was just thinking about
and then what am i trying to do
now there's an interest rather than what
am i trying to perceive
which is kind of interesting how they
went for that because
i mean i'm just thinking about there
could be that time for instance i'm
perceiving someone's i know saying i'm
perceiving someone's got a bit of food
in their teeth right
and what i'm trying to do is listen to
what they're saying
but i'm seeing that bit of food it's
kind of hard now
so then what am i paying attention to
but what am i trying to do is an
interesting one because in some ways
you could say what am i trying to do
i think implicitly saying what am i

trying to do with my sense oral
apparatus but it's a bit longer
um rather than what am i trying to do
which which could be a higher policy
level
thing yep
great question i think one possible
answer and this relates to potentially
the slightly different way that active
inference partitions
problems than other domains is
your preference
would be that the other person doesn't
have your preference at the first level
would be to see no food in the teeth so
there's two states in that preference
vector um in the in the c vector which
isn't going to be shown here but the
preference would be to not see the food
and then the policy
of the first perceptive layer is ocular
motor
the policy is what can be controlled at
that first level
so you could avert your eyes as a policy
that's one way
of not getting the visual stimuli
however
you can still hold it in your memory
that it is likely to be that case so
that's an example of like a hidden state
which is the state of the food all
you're going to get is photons
corresponding to whether there might be
food there or not and then you're doing
hidden state inference now even though
it's a visual so it's like why is it a
hidden state if it's visual because it's

actually your onboard state estimate
of something that's out there
you're
seeing it through visual mechanisms but
at the first level
the policy is for how you can move your
eyes
not
some other higher level policy yes
stephen
yeah that's that's quite a good uh way
of thinking about it and i think it also
shows
like how it
is probably more easily tractable when
you say you have the visual you have
something fairly clearly
on the visual
um or a particular sensory
modality um then you could see how the
attention shifts now obviously in in our
in much of our life but maybe not so
much in meditation because we we are
able to attune to
certain sensory
uh roots
more clearly but in daily life
it's all mashed together so much that it
probably is part of the reason why it
becomes imperceivable because
you know your body might be
in the mode of
the bigger goal are you getting to work
on time
um and then your visual thing in that
real time is trying to listen to the
conversation with the ticket collector
who's got a bit of two food in their

mouth in their tooth or whatever so you
and it would all become hard to make
this tractable you know because this
this isn't necessarily set up to think
about multi
um sensory integration not that it can't
but that would start to distort this
it's probably easier to think about this
where is a fairly clear
modality at play
another thing to know is we don't
necessarily talk about goals in the
active inference ontology we have
preferences
which are over sensory outcomes and
policies which are action capacities
action sequences that we're choosing so
goals are
they're a word and people know what
we're talking about but in this model
there is no goal for anything they're
just preferences and policies so i want
to
read this question
which is going to be interleaved with a
definition and then it will be an
interesting question
so liberty wrote
still reading through the paper aren't
we all
which seems deeply complementary to
metzinger's account of m autonomy so i
just want to read a definition of m
autonomy since it wasn't a term i was
very familiar with and this is um a
a paragraph written by christoph delega
who is also in the active inference
community so kind of

thanks for going out in front helping us
be clear about our definitions and so
what christopher wrote was
in his recent article metzinger 2015
argues for a new construal of mental or
m autonomy
that's the m as a functional property
which consists in the deployment of a
special kind of model one which
represents the self as an epistemic
agent
this epistemic self model is and then
here's the quote from metzinger
a global model of the cognitive system
as an entity that actively constructs
sustains and controls knowledge
relations to the world and itself
metzinger 2015 page 272
so
good call liberty definitely we're
hearing words like epistemic and
self-modeling and active states that
seem to bring us close to active
inference and so their question was
how do processes such as intentional
inhibition
inform
a formal account of cognitive control in
your opinion
so we can look at five while this is up
and people are thinking or
just let me know which slide to go to so
steven first or blue you want to go
just reread the question one more time
how do processes such as intentional
inhibition inform a formal account of
cognitive control in your opinion
so stephen and then anyone else who

wants to go for it
um so intentional inhibition
um or unintentional
uh
uninhibited
behavior or inhibited behavior i suppose
you could have it in different ways um
i think what for me what's interesting
with this when you start getting into
um
how we relate and how we fit into our
social
landscape so to speak with that type of
thing
is
when we because inhibition tends to be
something you're thinking about in a
social context within reason
it's like that thing about dances if
you're on your own
um
i feel that is a really good case for
where there'll be an intern locking
model such as the type of work used in
mental space psychology to mediate
um this metzinger
um
uh
knowledge
structure
because that then gives you a kind of
an embodied inactive
framework
which the brain can actually
use as a kind of meta model in
peripersonal space
um so i think that this is a quite a
good

point here is at what point
um
do we need to have a mediating
embodied um
sort of
metaphor or structuring to help some of
this
pass out into the complex domains of a
social setting
thanks steven
dean yeah i just think it's interesting
that if you
if you think about inhibition and you
think what the effects of that are
you can actually control the situation
if you don't talk because we all hate a
void or a vacuum and so by not
expressing
sometimes you you pull other people's
expression into
more than you maybe even anticipate and
so i
actually
have found
people who are
less extroverted a little bit more
introverted tend to have a lot more
power in these kind of co-variational
situations that maybe they believe they
possess
i know the people that are extroverted
think that they're controlling the
conversation but
just stare blankly back at somebody and
really pay attention and listen
and you'll be amazed at how much control
you actually possess in that in that
relationship

i have a thought on that but blue first
go ahead mine's unrelated it just
reminded me of the sort of street hustle
dynamic which is on one hand that person
must be
superficially extroverted i mean they're
coming up and speaking to a stranger if
it's about a chess game or they're
selling something or they want you to go
to that club in las vegas
that is a very extraverted social
activity
but the art and the science of that
straight game is actually about what
dean just wrote or said i guess like the
introversion that brings
the other person and elicits them
to feel like they should put out their
personal information into a void but if
someone's just like chat chat chat chat
chat
there's no space for them to become
vulnerable to feel like they're part of
the conversation and ultimately to
reveal personal information so it it
shows that um
of course we're using the introvert and
extrovert little loosely people say oh
it means they draw their rejuvenating
energy from being alone or in public
like there's different definitions but
it just shows that actually it's not
just about
saying something or being the first to
speak
that can also be a superficial mode of
control and that there's a deeper mode
of control

that actually has to do with picking and
choosing your words as well as when to
use them
blue anything on that
so unrelated but just going back to
inhibition stephen do you want do you
have something directly related
um well one little thing just just a as
a comment is one thing i remember
when they had the the press inquiry in
london about the levinston inquiry and
rupert maxwell had to give evidence
about his newspaper empire one thing
that was really noticeable and they were
saying this because he was like you know
he's a very obvious he's a high level
executive etc he always took his time
like when he was asked a question it was
almost like he was always used to having
the time
to think through his answer
say what he needs to do and be able to
reply his own leisure
in a way that other people might not and
that that also speaks to that power of
when someone who's presumably quite
dominant and be able to be forceful also
can they know they can hold the space
and there's probably some element of
knowing how much power you have to be
able to get away with that and no one
else steps in so that's quite
interesting anyway thanks
blue then dean
go ahead dean you've got some retort so
i'm gonna go back to inhibition when you
guys are done yeah yeah no i i i think
it's just really interesting because

i've read some stuff about
if you're an fbi interrogator one of the
tactics that's actually trained into fbi
interrogators is to be
intentionally inhibited
interesting so there's
intentional in the social intentionality
and then i think
could be
otherwise but it seems like metzinger's
account and and liberty's question
is about the intentional inhibition
within the agent as a mechanism of
cognitive control but it brings us right
back to this idea of the vertical
nesting
and then in that situation intentional
control
is kind of cascading from the top
but then also there's intentionality
laterally like i
am intending to control somebody else's
cognitive state
and so that's actually one reason why
it's important to have a model that's
kind of like legos in two dimensions
because these active inference models
they plug in really well
nesting
internally
so we can talk about top-down cognitive
control and maybe have intentional
inhibition as a phenomena within a
broader science of mental action which
we can get to in 12
but also
we can just as fluidly use some of the
same variables and some of the same code

some of the same formalisms for the
kinds of intentional control that
happened laterally so it's like when
somebody is taking a deliberate pause or
somebody asks a really hard question and
then they go
they take their time with the response
that's an intentional cognitive control
which has traditionally been the realm
of the sort of psychology perhaps but
then that can be understood to be
deployed
in a cultural context by which it's
being used to intentionally control
others states and so we can have a bunch
of different agents each consisting of
nested
models like this
and we can
stack them up vertically as well as have
interactions laterally because
intentionality is deployed in both of
those directions stephen
yeah and following on to what you're
saying this this then starts to
there's a big gap in
basically between psychology and
sociology
which is this idea or i suppose you
could in many cases teams is one area
where it gets looked at you know is
you know it's great you can take an
average and you can take your
statistical averages you get your
sociology
within in many cases and you get your
individual
but um

part of the reason that we end up there
is
not because they're the most useful
places to be is because you can measure
them you can get units of analysis which
you can
do stuff with
and the inter-subjective stuff around
four or five people is is a huge
challenge right because
you can't ignore the complexities of the
individual yet you've got the
complexities of the group
and
so that speaks to to that you know that
ability to now start to get in and say
okay well
let's say we can tease out some
generative models or
not even generative models i don't think
but to be able to tease things out a way
which maybe can make meaning in group
context and this participatory sense
making that's that's really what is is
moving things forward i think
it almost makes me think of
someone who can facilitate a group and
it's like i know you intended to just
share what you're excited about but
you're unintentionally exerting
cognitive control on other people
by
bringing up this topic or by speaking in
this way so it helps us look at that
vertical intentionality and the lateral
social intentionality together
maybe even
not just with the same model but on a

common footing
of precision
precision within the individual and
precision of the team so it's like these
are the real
nuts and bolts of where integrative
models matter if we had a totally
disjoint model for what was happening
inside of a head and then
across
bodies and brains
we wouldn't have um
the market to sort of flip back and
forth between those currencies per se we
would need another third level connector
theory so it's like if you have one
theory in theory b you need to invent
theory c in the middle but if you have
just one integrative theory then
it's just a question of how it all works
together dean
yeah just one last quick comment i think
it's really interesting that uh
that an inhibition
can be seen as a as a as a tactical
stance and a reveal on those markov
blanket hidden states
i mean that's
i'm not sure it's a cipher because you
don't actually know what the other
person if they're going to sort of be
drawn into something that they otherwise
wouldn't but i just think it's really
interesting that
we
we assume
that we have to keep the conversation
going

and we don't assume that maybe one of
the ways to
to do that is to not say very much
and to give one neurological comment on
that
perhaps back in the day people thought
with the higher orders of the brain the
higher thinking processes being the
generative and the creative processes
since those are the human specific ones
and were generative and creative and so
on but when you look down to the synapse
level
the frontal cortexes are actually
using inhibitory control
so it's this model where it's like we
can actually
be creative and we can be generative
through the appropriate
inhibition we don't need to have this
passive midbrain that gets exalted into
activity by a really smart forebrain
rather we can actually have the
endogenous wisdom and activity that
never seizes
with targeted inhibition
coming from
some other parts of the brain
i'm sure
that was a simplification blue so i know
you could say more on that
no so that totally like leads into
exactly what i was wanting to say
earlier about inhibition and cognitive
control so like if we were in this
overly
neurologically like neuron to neuron
excited state i mean that's a seizure

right so when you're you've got over
excitation in the brain it's it's not
always good but but so i mean to what
intent like to what extent does
intentional inhibition right this is
going back to the question that's what
they said intentionally
so when you have intentional inhibition
i wonder and this goes back to what i
was saying about meditation like
shutting off the fundamental unit of a
thought so i wonder if intentional
inhibition when you refrain from
blurting something out or when when you
practice restraint or or you know
refrain from letting your thoughts
wander so when you inhibit yourself this
intentional inhibition i don't know but
it's curious to think about does this
correspond to you know there's activity
in the brain excitatory makes more
action potentials happen and inhibitory
makes fewer action potentials happen so
when i do this kind of intentional
inhibition
does that inhibit my neurons in a
corresponding way i'm curious i mean i
would guess because it it just you know
i mean at least in the meditative sense
i would think inhibition of an action
potential propagating forward
so it's a curious thing to to wonder
about
to what extent does intentional
inhibition exert cognitive control at
the molecular level
[Music]
thanks blue steven

yeah that's
actually got me thinking now on sunny
that's quite
um
that
that question about the higher level
lower level i think and this this is a
really interesting piece oh actually and
it ties back to what i was thinking of
asking as well about figure 5 where you
put subliminal
processes and maybe conscious processes
i think that's an interesting
point on figure 5 that you've there
because maybe it's the middle bit that's
more conscious at certain times
and the top bit is actually never quite
aware of you're only able to effectively
feel
and the you know so there might be that
it's actually the middle layer there on
figure five that's conscious
um and i i like what that's actually
really interesting if it's inhibitory
because
there's an interesting model i i've been
trying to work with called dilts logical
levels um
um and you've got the kind of the
identity level which i'm if i map this
onto the the identity level or the face
um
could be seen as what broadcasts um
where we are
and
some of that we're conscious of
and some of that we're not you know so
and then we've got where our awareness

is in the moment and then we talked
about this kind of
lower level which you can only you're
sort of aware of but it's again it's
slightly outside of that crystal clear
um question so i wonder one figure five
and maybe this is a question for daniel
because it might be put that i don't
know if you put those on there
you know where you put we've got
subliminal and awareness and
um how is this interface mediated and
i'm kind of wondering what i'm paying
attention to maybe what is actually
awareness in
a
like sort of i could tell you sense and
the top one is an awareness in a kind of
a
fail almost like a metaphor it feels
like this to try and
have that meta awareness but you can
never quite get a handle on it you know
great point and
totally open to anyone's insights or the
authors or otherwise
but this first level is again about
sensory perception
so
it is possible that the second level
is the reportable opaque state that's
one of the innovations of this paper
and there would have to be another level
to make this state at the very top
reportable so it might be the case that
this green
topmost layer
is the felt awareness of

where one's attention is
as well as their preferences for where
their attention should be which might be
scaffolded by meditation saying
it's awesome to focus on this mandala or
it's awesome to focus on compassion so
these are shaping preferences
over policies
for how aware one should be it's great
to be paying attention and then the what
is one paying attention to is where
those policies are uh influencing yeah
where the attention is being paid but
when somebody says like you know there
was a car crash and i couldn't look away
it's almost like well the first layer is
that they were perceiving
cars on the road
the attention
they're reporting is that
they were paying attention to that
but then there's this implicit third
state that they were aware of where
their attention was
otherwise they wouldn't have been able
to report i couldn't have looked away
so they were extremely aware of where
their awareness was
that's what metacognition brings in
that simply attention doesn't
and
again another argument for first
principles theories that work really
well in this sort of composable way
because there are
neural network models with so-called
attention but where would you put
metacognition in there might be a

variety of ways to do that and maybe
some would be practical and some
wouldn't and maybe some would be
implementable with the tools you already
have maybe others wouldn't but here when
we can just look at the skeleton of the
model
we can
come to some pretty interesting places
just by working with what's composable
rather than just sort of trying to make
it um
fit well first
and then make it composable in
understandable seconds
we're beginning with composability and
comprehensibility and then the question
about how it will fit empirical data
well it's an empirical question
stephen
yeah this is interesting you can bring
in empirical data you've got that
advantage of potentially
say the subliminal layer it may not be
self-reports it could be that you could
use other
measures you know brain scanning and
other things to give a sense of when you
know what is coming in there
and of course as they mentioned because
attention
and again this is where we get caught in
our own need to make sure the literature
can hold together you know attention as
as was mentioned by lars is kind of
relatively uncontroversial right but uh
of course you get we we know from
phenomena you know you you sort of can't

help at some level get into the idea of
consciousness right because
conscious awareness
of attention is that kind of interesting
thing so when it goes to what am i
paying attention to and what am i
conscious of and then you get into
phenomenological phenomenological
consciousness or access consciousness
coming into play
um and it gets a bit fuzzier
so um but i think
at some level you you you you're gonna
start to hit those
um
in between those two you're sort of
hitting that kind of fuzziness of
where that sort of goes and then the
next level up again
um
probably is
more maybe it's actually more clear from
the next level up when you go from
uh attention to meta awareness of
attention but i think the the bit this
might be a
whole shift in ontology even could be
that shift from
how you're
perceiving or
and how that translates into an
attentional state
yep so in to kind of bring it to
multi-level or multi-level and
multi-sensory integration
the perception can be of two different
kinds like what gets passed to the agent
at each time point might be a visual

measurement
and a temperature or proprioception and
then
the policy
what am i paying attention to
could be reflected by paying attention
to for example visual input as opposed
to some other type of input
and then one could be aware or not
of whether they're paying attention to
the smell or to the sound or to the
light
so
how these models get integrated with
then
non-first personal measurements that's
kind of what we explored in this future
phenomenology slide with like okay where
do we bring in
empirical
measurements
and we kind of align them on the same
time series and we can bring in
any kind of measurements we don't even
need to
um
limit ourself anything that's measured
through time of any uncertainty over
that measurement we can kind of cobble
it together into this model
what are we doing it for
what do we want to know with those
measurements
probably a lot of things that we could
address there
maybe for the
45 minutes let's at least go to 12
figure 12

yeah and
talk a little bit about
just annotating or exploring
this
figure
so in five
again main contribution of the paper one
of them
was the explicit three-level model with
um
what is being perceived what is
attention being paid to and how aware is
one of that
here we're dropping back to two layers
with the understanding that we could add
another level like a green layer on top
of the orange one that then connects
back down to the orange one
but look at some of these new parameters
that are introduced
in this paper
um namely all of these gamma subscripts
so gamma are the uncertainty variables
now there's a gamma with a subscript
modulating
a lot of the
other parameters
that previously have been presented just
as kind of standing alone so like for
example
in um
figure five you
just have like preferences in the second
level the preferences just exist and the
affordances e they just exist
now in 12 we're thinking about what does
it look like to have an uncertainty or a
precision

being the
inverses of each other
over
d
the prior
c
the preferences
g
the free energy minimization
e
the affordances
pi is the policy
and then we have an uncertainty over b
which is the matrix that defines the
transition between hidden states
so like given that i'm inferring it to
be raining outside what is the
transition probabilities of it staying
rainy versus becoming
not raining
and then we also have a which is the
matrix mapping between hidden states and
observed states what's the relationship
between the observation of photons and
whether it's sunny or not is it a simple
matrix just a clean mapping you can just
sort of simply have a hundred percent um
of the time map it or
is it the case that you're very
uncertain about how hidden states are
connected to observable outcomes
so those are just just to sort of recap
what these parameters in a model mean
but maybe if anybody wants to like
highlight one of these
precision variables
just what's one that they're drawn to or
they're curious about or

what's one that might have applications
in different systems
daniel i i would speak to it but i
in order for me to sort of explain it in
a way that i think would make sense
might mean i
would have to speak for a while and
hijack this would people be comfortable
with me talking for a couple of minutes
please
okay
so
i kind of look at this from the
perspective of when in doubt zoom zoom
in zoom out in order to compute
something and this is a computational
paper um
i kind of take a whole cloth or a whole
markup blanket approach
of precise or precision as the paper
kind of does a good job of laying out as
precise
ie figure 12.
so on the topic of meta awareness
meaning and collective decision making
between the greatest and the smallest
distances that we can be aware of we
have information geography
if you're searching for meaning and edge
precision things like contrast and
proportion
as a fitting process i often think of
like the 16th century maps of north
america when the explorers
first hit the coastlines
and then there's information geometry
when you're putting a dimensional space
together so that's fixing the topology

as a precise representation
of difference over distances
i've already shared in the 21 or 28 one
thought i've got this generative model i
call it the unholy triad which is in
essence a computational phenomenon
phenomenology
context generator
that translates uh prediction error
reduction meaning we try to give people
a sense of prediction manage management
expertise
i want them to be able to turn that that
ability to predict into a contribution
for me it was trying to create a
professional contribution for high
school kids
so what does that mean it means getting
to the geo the geometric and the
geographic essences of a situation
that you've never been in before so it's
a pretty big markov blanket inactively
it means generalized and transfer
regardless of that degree of novelty
it means understanding learner
availability to phenomena as much as
collecting data on learner attention
which is what this paper is kind of
focused in on it's not about studying
the swallow and regurgitating
information
as a content cycle
which is a standardized form of learning
measure which is also an irony because i
had to take these models here and give
them just 16 to 18 year olds by using
things like when in doubt zoom in zoom
out because they i could present them

with this but this would be
sanskrit to them they wouldn't
understand it at all
[Music]
so then what is a precise proposition
as create a professional contribution in
that to be materialized or propositional
sense and in the paper
i think i talked about this in 281 there
was a passage in the paper that talked
about
and i'll just read it out it says in
computational terms the central place of
confidence
and reliability and effective
self-regulation speaks
to the key role of precision
and then they say that's the inverse
variance of implicit beliefs
and then they talk about
as an example probabilistic or
posterior-basing beliefs
that describe a probability distribution
over some latent quantity
and i talked
last week about
going across the street
so there's a calculable risk in that
and then back to the paper it says these
beliefs are subpersonal meaning that
they're unplanned however the argument
pursued in this work is that basic
beliefs about beijing beliefs can
in certain situations
become propositional and and the way
that those things are estimated
optimized or controlled that's referring
to hesps casper's work

precisions in a nutshell quantify our
confidence and our beliefs say how
confident we are about what we know
about states
of the world about their relation to our
sensory observations or about how these
states change over time by analogy of a
physical action a mental or covert
action consists in deploying and
adjusting these quantities
without there necessarily being an
explicit behavioral counterpart to these
covert actions
so for me
if we're computing a transformation of
unplannable to metacognitive from
perception of
i'm aware of planning to make a plan
because i can think about my thinking so
what of this policy of attention
then what i attend to tells me what when
i'm losing focus
that's what the paper said and i agree
check
that's one way of measuring that and
in the crosswalk case study you have two
propositions both are precise in terms
of their
their dissem disambiguated paths
and both are examples of a nesting study
in this case crossing a nest or a
roadway
and then what happens regardless of the
awareness level inside or outside of the
physical and the hypothetical lines this
is what i would be asking people that
were going to go into these novel
situations if we don't select right away

if we keep things open
if we keep probability distributed
across a propositional horizon of
potential street crossings is precision
lost as an openness
as a district as a distribution
collection versus a focal point
attenuation which is i think what this
paper was examining
perhaps a way of probing this might be
to say that the concept of scale is
fluid
that it's the potential for scale change
that scale itself overtakes or retreats
from as a directional inflation
or deflation
and sometimes we talk about that as a
directional gradient to center descent
in this sense scale either obliterates
or gap increases which scott talked
about having an epoxy
and i agree with that
that what we consider to be inside or
outside the lines is not static due to
our relationship our betweenness to the
focal point of interest
this is somewhat
this is a somewhat conventional
explanation of scalable behavior but i
think there's a second part to this we
have to
keep with that first and that second
sense
which together with the first part
generates another type of between
which is a matter of the geography in as
in edge precision
and geometry out as in of difference

over
at once at this exact same time
that supports the very notion of and has
scale scaling down or scaling up
so that's that up and down part we were
talking about earlier today in my
experience is very much the addition of
this second sense
as a parallel probing that works there's
no violations
of the bringing in the statistical and
the physical mechanics together
that active inference parameterizes as
an hov or high occupancy vehicle
lane as a metaphor more
than intersectional
intersectional is pretty much
interventionist
and it's kind of an instructionalist
only reinforcing
view or perspective
in the in the hov lane metaphor
you are in
as self-organizing
your non-interventionist because you're
a constructor you're an interactor of
line live stream lines
and i use the example of a 28 0 or a 28
one in the way that we set the live
streams up
if you're outside the hov lines you may
be you're re-watching
something that you were a participant in
or
perhaps you're using the 0.2 part of
this
as a way of sort of detaching yourself
and being able to appreciate scale

so math is not the territory but the
math is both the diamond shape on and
above
the parameterized state space that road
that we now say is high occupancy
vehicle it shows the signals
meaning the geometrics and the
restricted fit the geographics
that is two plus in the car mathematical
quantity enables
again you have opaque passengers
and transparent traffic congestion
reduction check
that's what this paper talked about
but i think we also have to consider
that the contrast meaning that the text
in the background are different colors
and the proportion meaning the depth of
field is adjustable
is necessary to dial in the resolution
and the resonation daniel you spoke
about resolution and resonation mostly
about the resolution and the point one
and so i wanted to really come back to
that a bit
this is collected phenomena this is not
i am attending or i realize that i'm not
attending but i think it's a big part of
this attenuation versus collective
thought piece that we're trying to
get some sense of order and relationship
around
if we want context generation the in
plus the out equaling context to play a
former role formal role
in what we decide or define sorry as
precision it's something to think about
when looking at logic as a formalism

the accessing of data collection sets
which we've talked about again quite a
bit today
uh talks about the definition of a
precise bigger picture as a distributed
collection
and then like i said in my work that was
coming up with a professional
contribution
it also speaks to where there are
relationships
through lines
like like a real physical line and those
blankets that we keep talking about as a
process
and that also talks about what scott
talked about in the la in the 21.1 is
borrowing from the model has both a
quantity and a quality determination
i think it at the end of the day it says
instead of just looking at how and why
we have to look at the when
we're inside and outside the lines the
who
our relationship to all of the uh
traffic that can be crossing the road
and the which
which
do we look at do we look at the figure
12 or do we zoom out and add a third
layer i don't think it really matters as
long as we're prepared to look at both
so that is that is in to me the essence
of what a free energy principle as
unplannable
to
cognitive metacognitive
world

looks like and i don't know that it's
easy for people to sort of put all of
that into a nutshell but like i said
it's kind of a whole cloth
whole markov blanket thing and as you
mentioned daniel in the last
in the last live stream it's it's we
want the first principles to be able to
generalize and scale up
but i think the only way that that
happens if we is if we don't see it
as either always inflating or always
deflating but we see both the geometry
and the geography sort of working hand
in glove with that i think that's how we
derive a fit here
nice dean
thank you
very interesting um
i guess steven or blue or anyone else
feel free to ask a question but just a
few notes on that
so one is it's a subtle design choice
but notice that the green wraps around
the whole box
and then the orange wraps around the
blue
so it's not just like three legos
placed one on top of another it actually
is a zooming in and out and so that
moves us towards bucky fuller's dream of
you know i went in stairs and outstairs
rather than upstairs and downstairs
because
it's not up and down for everyone just
say it's in and out because you're on a
closed surface so
that was about zooming in and out and

it's not that we're even necessarily
moving again we talk about top down and
bottom up and that's vertically how it's
arranged here but perhaps
um inward out and outward in will make a
lot more sense when we start to think
about what these models are really doing
and um you brought up the idea of using
the dot two and so that was interesting
to think about resources and artifacts
that might be considered epistemic i
mean it's research we're talking about
papers but how do we use
resources rather than just learn from
them so seeing them as action artifacts
rather than just epistemic artifacts
and then also just this idea that
where
will we have precision can we have
precision in our policies
how do we
instruct a high schooler or younger
their world that they're going to be
having a career in and that's what it
sounded like you were working towards
was some sort of you know helping them
get on a trajectory where they could
have support and everything
it's going to be changing a lot
so
it's like i'm sorry i can't give you
precision on
whether crypto will be important in 15
years or something like that it's like
we could write a million papers on that
and just not know the answer today
what can we have precision on
and then this really pushes the question

um
where can we exchange our uncertainty
and where what what should we focus on
so that we can have focus on if not our
own policy like i'm gonna go on a walk
every day but our own attention
and that kind of covert mental action um
to bring it back to intentionality might
be
what we will always remain sovereign
over is our attention
stephen
yeah i think
the
the this this nestedness is important
and i think of course time is part of
that as well because these lower levels
happen quicker
and i wonder dean what your thoughts are
on this um
in terms of
when we look at these different
variables that can now come into play on
figure 12.
i mean they're kind of okay you're going
to start to have
variational potential to vary
how you set because in theory that every
one of these values is is there
implicitly just can can't be varied
right it's like there's there's there's
some sort of precision somehow in the
system but theoretically now okay you
can
access or change the precision on your
prize and your generative model on your
expected free energy
as well as on your transition matrix

that gets you the sensory states and of
course then this question is okay now
it's showing this in one
level
all these arrows come together somehow
it's like that mystery well how does
that happen right like is it a case that
they're all
firing out with like okay well this is
what it would be like if
um
you know
my updating of states was to have a
different precision this is what it
would be like if my generative model was
to have a different precision this is
and is there almost like a a whole array
like imagine going in and out of the
screen here of temporal depths or
even
different sort of combinations of these
uh
you know precision estimates and maybe
the top level has to basically choose
and and maybe at some point the lower
levels if they don't get chosen very
often they die down you know i.e like
the
seven legged unicorns running across the
street isn't selected very often so it's
like left to one side
um but maybe with creative people they
tend to like keep all those things
firing which may not be healthy if you
go too far with that
um so i'm because there is there is a
question there isn't there is how
as we start to bring in these other

ideas is okay so what what what's that
what's that ability to to bring up
a precision estimate on an a matrix
for looking at observations into states
and to integrate these with these others
you know and
is that is that just
you know integration you know are we
just gonna start to have kinds of
things like you're saying if you if you
have a particular change in in how much
the priors are changed
um
and along with the model you know then
what does that do
yeah i actually have some some thoughts
on that stephen it's interesting because
i think all of these figures and every
paper has their figures because they
show their model i think that that model
is kind of a reflection
and i think the other thing that tends
to happen is if you if you do reflect on
so what what was i trying to perceive
in terms of sort of a metacognitive
here's now i'm now i'm thinking about a
process
i don't think we do that in isolation i
think the environment kind of acts as a
scrutinizer as well like you guys are
scrutinizing what i'm saying right now
and i'm hoping that i'm not saying
something that's so completely obscure
that it it sort of continues to build on
what we're all sharing in terms of what
we're taking away from this so i think
there's an updating of updating and
that's when

when um
lars was talking about that in the 28
one i was kind of nodding because yeah
that's
i think that's a that's a vital
component it's not
sort of written out explicitly in this
model here but i think
we should assume
that when we're zooming in and zooming
out
we are reflecting
so we're taking something and we're
stretching that time frame
and then the thing i think the second
thing that we can't sort of leave out if
we're context generalizing is that
context that environment is acting as a
scrutinizer it's giving us feedback all
the time
so that we don't get stuck
in one place that time actually and
change actually does
becomes something that we're not just a
we're aware of but we're actually
utilizing
so that rate of change pieces i think
we don't necessarily impose it i think
it gets imposed on us or maybe i'm wrong
but
yeah i haven't seen any evidence to the
contrary yet
um actually that's that's a good point
if i just jumped in those is
yeah i mean a lot of this stuff
could i mean it actually starts to
support the inactive approaches in a way
i suppose

you actually start to see now that
you're back into the realm of well how
could you
start to blend all of that it almost
needs to be maybe hardwired into
an active kind of
dynamical biological
thing
the technical term um so
because
it's not necessarily tractable
purely however there may be parts of it
which need to be more cognitively
uh and can only happen at a sort of a
more structured cognitive
level you know so
there's this question about what what
can be inferred
through acting in the world in different
ways and like you say the
you will find out when you bang into the
wall or when you or you just find out as
you drink a cup you know you feel that
you're feeling
you know hamilton's least action
obviously becomes one of the options
there is that would come in with a free
energy okay
what seems to be the
type of configuration of precision
estimates that kind of gives the overall
least action
um
if that's if that information is
available
and so now you have a
way to integrate those
um in one context which may provide the

general structure which is good enough
to to to work with them in other
contexts you know maybe that there's
some sort of
plausible
so it sort of moves beyond neuro
phenomenology
and moves a bit beyond the body and
you're now in you are in the mind body
environment dynamic you know
yeah i think the really interesting part
is
when in doubt implies
you're you're you're stuck
you're not maybe taking action
so zoom in zoom out like don't just do
one or the other
be prepared
uh and actively but also from an active
inference perspective to
infer the geometry and the geography
that and i mean make that explicit
make that make that a part of your
strategy your tactics it's actually
it's more inhibited than most people
think
but in terms of in terms of what you're
going to do is you're not going to get
stuck on
level one
or level three you're going to actually
be able to go across levels that's why
there's an envelopment in a lot of these
representations you're already in the
nest you're already deciding whether or
not you're going to cross the street now
what you're asking is is it more
beneficial from a thinking perspective

to that insider outside the lines i'm
not i'm not passing judgment on which is
better but what i am saying is don't
restrict yourself to one or the other
zoom in zoom out
the opacity of the second layer
only arises because there is a higher
and a lower
level
right
so it it's sort of like
maybe the series kind of blurs off and
so we have one type of awareness at our
experienced tear
and then it rests upon the shoulders
of the first and the third level the
green and the blue here
and then maybe those spiral off in
reality or maybe they don't
we're modeling it this way
and then it's the feedback with reality
that we'll update our models based upon
stephen
yeah
and
this this because we've got levels at
play and i know you you mentioned there
about zoom in zoom out i mean zoom's
quite an ocular kind of
so i suppose we've got
there's a there's a kind of a question
in terms of how we language some of this
because
we we want to look at different temporal
we've got different levels of
abstraction as we see when we then look
at things as they map across to our
other knowledge epistemic systems we've

got different we know that the
hierarchies
have different speeds i mean in a way
they don't
nature in theory doesn't care about the
levels in sense but it might care about
the
the the speed at which the processing
happens and if it has to go quicker at
the lower levels it has no choice around
that
and that but then we've also got this
question of zooming out scale-ups you
know scale
scale happening some of which is is is
you can't avoid and we've got cells
inside our body a set of scale that's
and then we've got organs and then we've
got
i'm wondering
yeah what
when we say sky zoom in
um is that an intentional
no okay i'll pause it no again no i just
that's a great question because as i
said i don't have any pushback on
anything the authors wrote in this paper
i think it's a great paper
and i'm i'm i'm suggesting we hold up
attention
and availability because that's what
zoom in zoom out is it's leaving
yourself available to what could be
considered
in included in a definition of what
attention is
i'm just i'm saying
include both

that's what the zoom in zoom out does it
says what am i what am i available to
am i avail am i making self available to
outside the lines metaphorically
physically
as well as inside the lines is that part
of my reasoning in terms of of not
necessarily be being afraid of bumping
into things
that's a good point i mean one other
thing that i think is interesting when i
think from
science to
visual art with art
is in science you tend to think of
zooming in and out say with a microscope
right because you can go in and you see
more
now in art you tend to crop
so you have the same roughly the same
image maybe you go a little bit more
detail but you're just cropping in
now cropping
in terms of how we trying to think
informationally is quite different right
because it's what you're taking away to
leave focus
we're zooming in because we get more
detail on something small
we maybe don't focus on all the things
that and well you could zoom in and you
you just
expand the you know
the frame gets bigger right
normally if it kind of that's the
question are you zooming in
on an object that's like a penny and it
gets to a certain size until it fills

your frame of awareness because once you
go beyond that you can't help but crop
because you can't fit everything in and
you then
gain more detail on something that you
can't
take in other detail
so there's there's um i mean even that's
a scale-dependent thing right it's
dependent on the organism right so
for a mouse to zoom in and zoom out um
well also can to some extent without
using a tool or an instrument can we
zoom
like
it's interesting because we
if
can i zoom with my eyes
in a way i can
focus and i can go and i can
take my
most accurate i suppose in some ways i
can zoom by having the area of my eye
that has the most
um
cells being given attention to a
particular part of the
i i can take my attentional
um
fidelity to bear
but some of the things also like i say
we we we kind of imagine ourselves as if
we've got a binoculars or some other
tool in our hands right and we um
because we sort of it's such an embodied
thing to do
but actually that's not necessarily
that's not necessarily what we could do

without that tool if that makes sense so
there's a there's a there's a question
there around
um
yeah
where
where do and maybe we were able to use
maybe that's the thing that's kind of
unique about humans is that ability to
pick up a microscope and look through it
and
extend through the tool usage you know
and
that's not necessarily available to
other animals
i think i think that
the paper speaks to
uh i can i can attend or not attend and
i'll know that i'm attending when i'm
not attending
so that's
that's that's the sort of inflationary
deflationary
part of it and i'm not saying drop that
i think that that's important but i
think if it's only held up as as that
you're missing the other two vertices of
the triad the other two vertices being
can i also on the on the open end of the
triangle relative to that first part
which is the
mir am i not attending am i inflating or
deflating am i going up or down
is the geometry and the geography and i
think if you put all i know this sounds
very abstract but after after you put it
into practice
after you

have your checklist that your or your
model that you carry in with saying have
i have i addressed all three sides of
this
you tend to generate a context much
sooner your mark off blanket
becomes more transparent more tunneling
effect into these novel situations
then
i'm waiting around for somebody to point
to
what i need to attend to next which is
more the of the instructionalist model
again
right
so i mean if you want to give a person
the ability to
make sense sooner
all i'm suggesting is is that have three
vertices
right have three corners to this
to dimensionalize the space and and
again if you put it into practice
it's it's it's pretty easy to transfer
from the abstract into actual
physical things that you do
i mean you you can you can decide what
the focal point is and what the horizon
is you can decide what the difference is
and you can decide what the
what the proportions and the contrast
are and if somebody says to you
i'm not getting this you can also
explain to them what you're not seeing
which is really really helpful to the
person who's trying to be the be the
guide of the wayfinder
thanks dean

blue
yeah i wanted to kind of talk about
zooming in and zooming out and um using
a microscope or not and so something
like that i have found in the course of
my life is
the capacity for mental zooming in and
zooming out um especially with regard to
um consideration for others so like i
started life as a kid like everything
was provided for me and then like as a
teenager i was like the most
self-absorbed creature alive right and
so like as i've gotten older like my so
i was so focused on just me right just
me me me me i'm the center of my world
but then like you know you really start
to expand that um
you know to consider more and more
people and the planet and like all of
the consequences of your actions as i
you know developed and you know probably
really set in at like 25 right with full
frontal lobe development and the
understanding of consequences and you
know this kind of ripple effects the
butterfly effect of our of our actions
and so forth but also in meditation to
kind of bring it back in and tie back to
the paper
um you know you hold you do this
meditation i mean i've done meditations
where you're you know doing for example
loving kindness and you know you focus
on loving kindness and it's kindness for
yourself and your inner circle and then
you expand it to include you know the
people in the room and your extended

family and you know everyone that's
suffering the same ailments as you and
so forth so you kind of expand your
awareness this mental zooming out
um through through meditation and um yes
so i just wanted to kind of say like we
don't always need a microscope but but
it is mental
and it may also be something that's
uniquely human
can i can i just respond to that blue i
think you're absolutely right and the
funny thing is is that
again when he when you take it off the
page
it's what you're describing what i'm
describing although i'm probably doing a
terrible job of it is identity and
feelings
when you actually use this this
its approach of being able to hold up
three things at once it's your identity
and your feelings that you're really
becoming metacognitive of
or so is it metacognitive or or is it
dissolving
the identity right and so that's like
the question of mike levin you know
expanding the cognitive boundary of the
self
so
in his paper i'm not sure if you read
the computational boundary of the self
or not um but but you know he talked
about this the computational boundary
and and you know not just of a person
but also like a cell and an organ and a
tissue and and so but also a human and

and when you've dissolved this
computational boundary of the self
you've expanded it you've dissolved i
mean this is eventually the end goal of
meditation is this dissolution of
identity this ceasing of grasping to the
self i mean in many traditions right
this is the letting go of self-grasping
and letting go of the idea of the self
and so you have this very
expansive
consideration of the self and so it's
encompassing all um
beings in that in that boundary yeah and
and that's interesting because that's
exactly what we used to talk about is
what does it mean to go from being a
student to being a professional
contributor well all i know is it's not
an extension of being a student
i have no idea what it means in your
particular situation right so it is a
dissolving
if if nothing else it's a forming and a
dissolving at the same time that's why i
said
you still you still have me because i
it was your influence that got me
watching soul
and that's exactly what they were
talking about and so again it's not a
it it is it is effemorable ephemeral and
we can reflect on it and we can have the
environment scrutinize on it and that
doesn't destroy everything that's
actually building things up
thanks dean stephen
and this this what you're just

mentioning there in terms of
dissolving the self
and meditation
and precision
estimates um i think this this could
speak to this idea you know we talk
about the five there's six plus or minus
one in terms of how many ideas you can
hold in your head and we have this idea
you know with higher thinking it's pure
or we have certain
things that we hold in our minds and i
can
imagine that it's like
with meditation the idea
if it's done well because it can you
know you could end up going into some
places where you just don't
it's not productive but in a productive
sense it's like you are able to adjust
and integrate you can imagine
y d y b y c y e
y g y a
and
b o it's not that you so you're still
accepting that you can go with different
precision estimates but you don't get
this kind of attachment
or
to making certain precision estimates as
as kind of rarefied things which we
often do you know like so we kind of
feel like it should be like something or
we feel uncomfortable or we get drawn to
our kind of um
you know
our desires or the things that we get
kind of obsessive with or things that we

are avoiding so that could be
interesting and the other thing that
then can tie to that because i like to
mix as well as meditation which this
talks about which is kind of yeah you've
got meditation you've got action in the
world
where we can like you say with the
environment walking around and then
we've got ritual
and ritual tends to be more in the um
indigenous approaches but there
you're not necessarily trying to
dissolve anything it's already
it's almost like you're trying to
disrupt it
it's not like you've got a nice
pure
drop that you're hoping to like
evaporate and allow to become a mist
that we're not rarefying anymore it's a
bit like you know you you see that you
you're going to put on a big mask and a
costume and you're going to dance around
the fire and until you start to
go into another state
now that could access another
set of
informational
inferences
um and that that may be doing some other
things you know that maybe things which
are more useful when we're in a communal
integration
and social integration or social sense
making
um so it could be like we're trying to
regulate as a group and then meditation

is helping us regulate as individuals
and it's neither one's better than the
other but they have different
you know niche
properties for us
or well interesting
discussion
maybe we can each give a closing thought
and then i have a short quote to close
based upon something dean said but
it's um
it's a really interesting paper i think
we all
enjoyed reading it it really does point
the way
towards uh
what does a generalized computational
phenomenology
or at the very least without getting
experience into the mix just what is a
computational framing of mental action
is that a paradox in and of itself is it
impossible is mental action just like um
a no-go
or is mental action going to be
understood
to be
everything of the iceberg that isn't the
tip of the iceberg
blue so
just a final thought um you know i i met
a this metacognitive aspect and you know
that the idea of identity dissolution
and i do think it was mentioned um in
this paper but just briefly like what
about under the influence of psychedelic
drugs where we know we have this
identity dissolution this ego

dissolution this expansion of the
cognitive boundary so so when you have
psychedelics how
is the metacognitive
hierarchy
influenced disrupted is it obliterated
is it enhanced so i'd be interested to
see if the authors go down that road in
the future or to see someone you know
take it to that level exploring the
metacognition of
people under the influence of
hallucinogenics
thanks blue dean
i think back in the 20 zeros we were
wondering about whether or not this
this paper and the stuff that it points
to could be used for nefarious purposes
and i
think i commented back then that
i used to work with a lot of people who
are spending a lot of time with focus
and attention on task
and i just like
to sort of keep that where that is
because i know that's a thing and i'd
like to just sort of hold it up with the
background of what we're available to
that availability piece because i think
that's the more powerful thing
and i'm not sure that that's necessarily
threatening but i think if we can have
conversations with people about
how active inference
acts as an availability enhancer
they might be more attracted to this and
not necessarily see it as something that
people could use for

for nasty purposes i mean there's going
to be people out there going to do their
nasty thing
but i think there's actually a really um
a really empathetic and sincere
aspect to this that
isn't just exploitable but is really
explorable and is really kind of cool
we'll have a quantitative framework for
saying that we would prefer if they
didn't even though we expect them to
behave the way that they're expecting
but um
we'll have an interaction sequence that
might change the way that that works
steven
yeah a couple of points well one it
might be interesting to see what adam
saffron makes of some of this with the
psychedelics as he's just started a
postdoc now at uh in psychedelics at uh
john hopkins so that'd be uh
that'd be interesting
and then i suppose the other thing that
i've noticed is as with some of the
nefarious actions and i think you're
right but i think also
what i've noticed is some of the kind of
intuitive nefarious actions of
politicians who are particularly adept
at
getting people on their side
it may actually be that some of that
they're doing through kind of like some
sort of roundabout intuitive artistic
route this might help us reverse
engineer how they're doing it so there
might also be some benefits to

to understanding what is it they're
doing even if they don't quite know how
to describe it
dean
just one
i think you're absolutely right stephen
i think
if we if we
are able to make this available to more
people their detectors are
going to get really really improved
so the closing quote to hopefully bring
some of these threads together we talked
a lot about you know distributed versus
centralized sense making and what is the
role of scrutiny and the scrutinizer
like dean brought up and of course geb
hofstetter's book from 1979 but another
work of art from 1979
was frank zappa's song central
scrutinizer and just closing with some
of the first lines of that poem
this is the central scrutinizer it is my
responsibility to enforce all the laws
that haven't been passed yet
it is also my responsibility to alert
each and every one of you to the
potential consequences of various
ordinary everyday activities you might
be performing which could eventually
lead to the death penalty or affect your
parents credit rating
thank you everybody though fun times see
you in